

CTS SRE – Leveraging SRE for Enhanced Service Lifecycle Reliability

EMPOWERING High Performance
Technology Teams

Agenda for the Day

Duration	Activity
5 mins	Welcome and Objectives Understanding the role of SRE in service lifecycle reliability.
10 mins	Introduction to Service Lifecycle Reliability Importance of reliability in development, deployment, and operations
20 mins	Driving Success with SRE Practices System reliability, scalability, and operational efficiency. Real-world examples and use cases.
10 mins	Break
15 mins	Benefits of SRE Implementation High uptime, cost optimization, and revenue growth. Interactive exercise: Identifying uptime metrics.
10 mins	Achieving Operational Excellence and Innovation Balancing innovation with long-term reliability. Discussion on SRE maturity models and post-mortems.
10 mins	Q&A



The Session Equations



Service Lifecycle Reliability

- **Overview:** The importance of reliability across service stages (development, deployment, and operations).
- **Key Concept:** Balancing reliability and innovation in a dynamic environment.
- **Example:** Highlight Google's approach to using SRE to manage large-scale, highly available systems.

Google's SRE approach

Product lifecycle

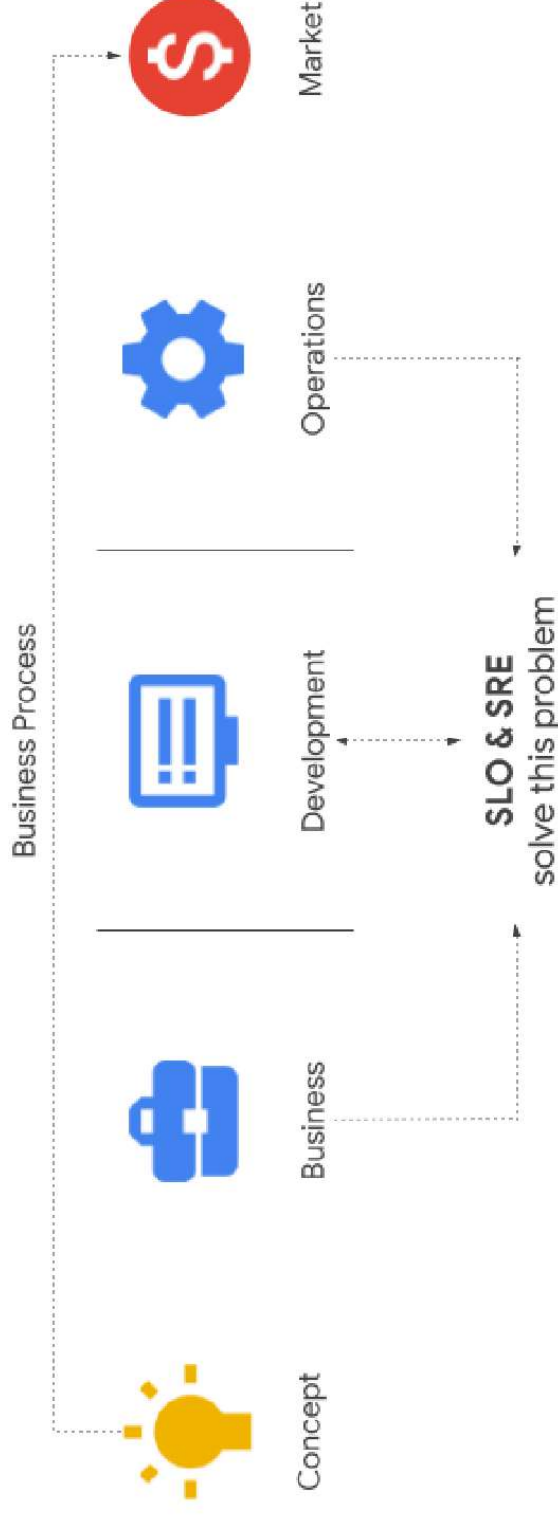


Image Source : <https://medium.com/@devdoot/4-site-reliability-engineering-sre-concepts-and-components-ofbb9e8296b3>

Business at the Speed of Thought



Present Paradigm



Proactive



Reactive



**Be a Air Traffic
Controller**



Aircraft Captain



Be a Gourmet Chef



**Stewards &
Waitress**



Be a Nutritionist



**Physician &
Surgeon**

Business Platforms Today

Fifteen practices help define a world-class software-development organization.

Setup decisions that guide the strategic road map

- 1 Cloud-migration path
- 2 Platform choice
- 3 Microservices/container architecture

Product-delivery practices to ensure quality delivery

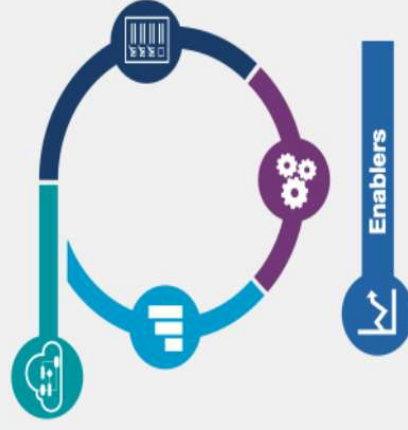
- 10 Analytics and use of telemetry
- 11 A/B testing
- 12 Community-driven development

Product-management practices to aid in product conceptualization and design

- 4 Product-management excellence
- 5 Human-centric design

Product-development practices to build and test quality solutions

- 6 DevOps (CI/CD¹)
- 7 Test automation and TDD²
- 8 API³-based architecture
- 9 Productivity and quality



Enabling elements to plan and operate

- 13 Portfolio management and product economics
- 14 Talent and governance
- 15 Product security and risk management

¹Continuous integration/continuous deployment.

²Test-driven development.

³Application programming interface.

McKinsey & Company

Image Source : <https://stage-geoffrey.be-web.be/Must-Know-Tips-To-Start-A-Software-Development-Company/5414696.html>

ENTERPRISE INVESTMENTS

Business Process Management

Architecture

Compliance

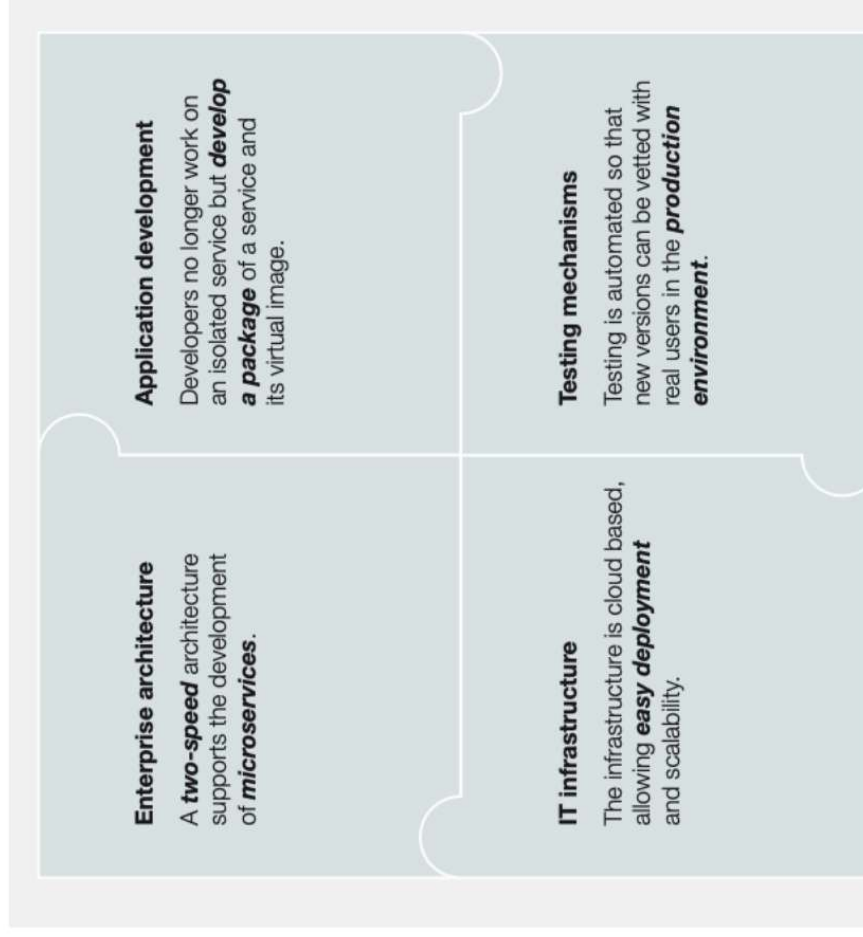
Development Practices

Virtualized IT Infrastructure

Continuous Testing

... NOW SRE !

Business Platforms Today



BUSINESS CONSIDERATIONS

Continuous Features Update

High Availability

High Performance

Highly Secure

Experience Economy Ready

Planet Scale

Evolution of Eco-Systems

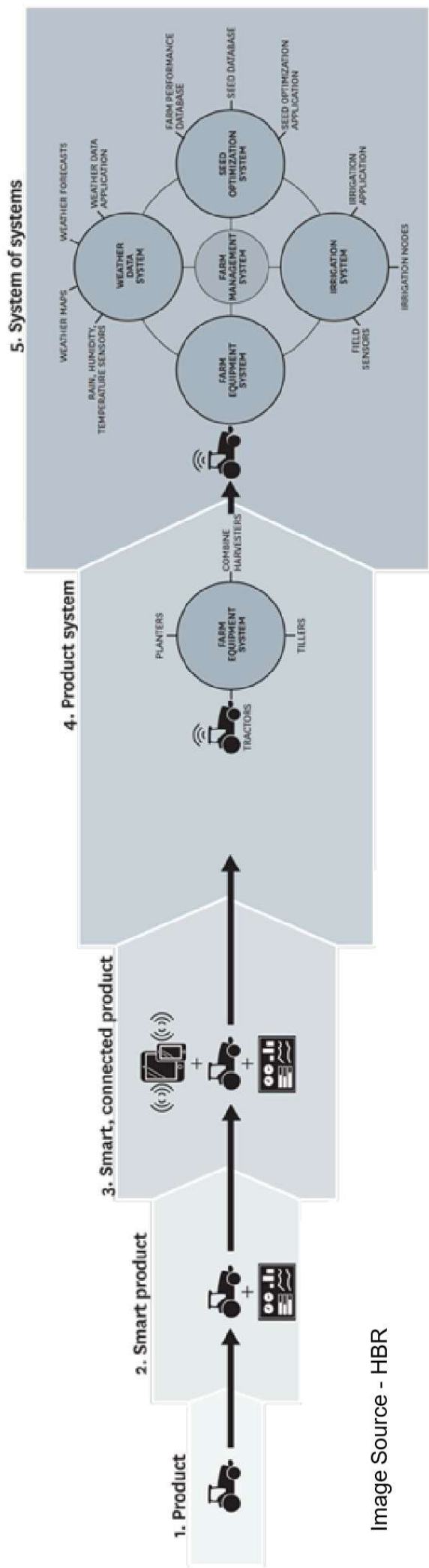


Image Source - HBR

Future World...

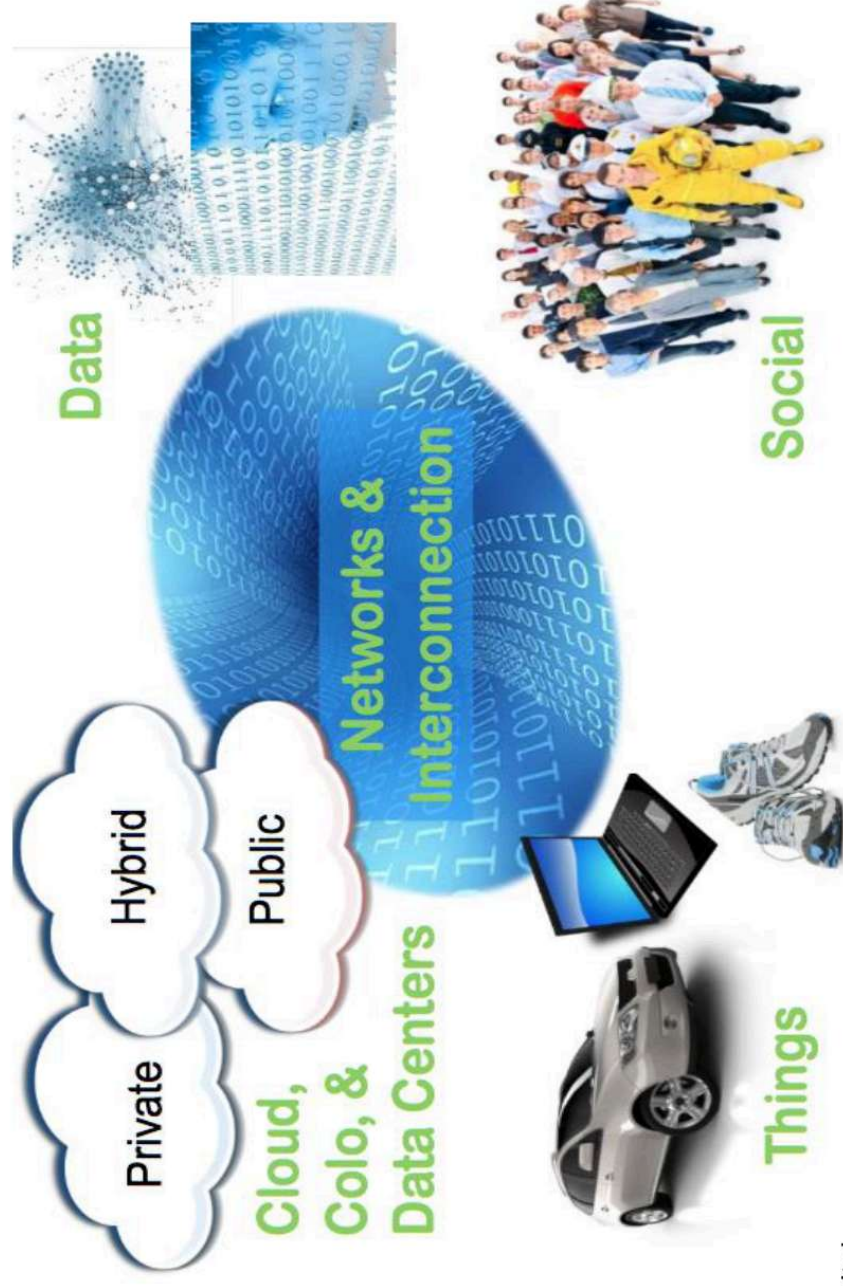


Image Source: Digital
Disciplines Book

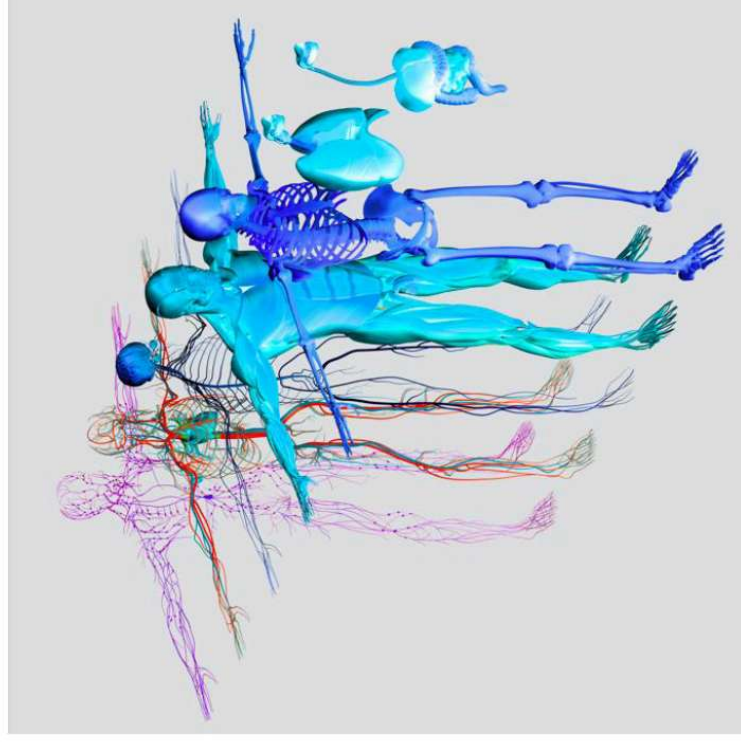
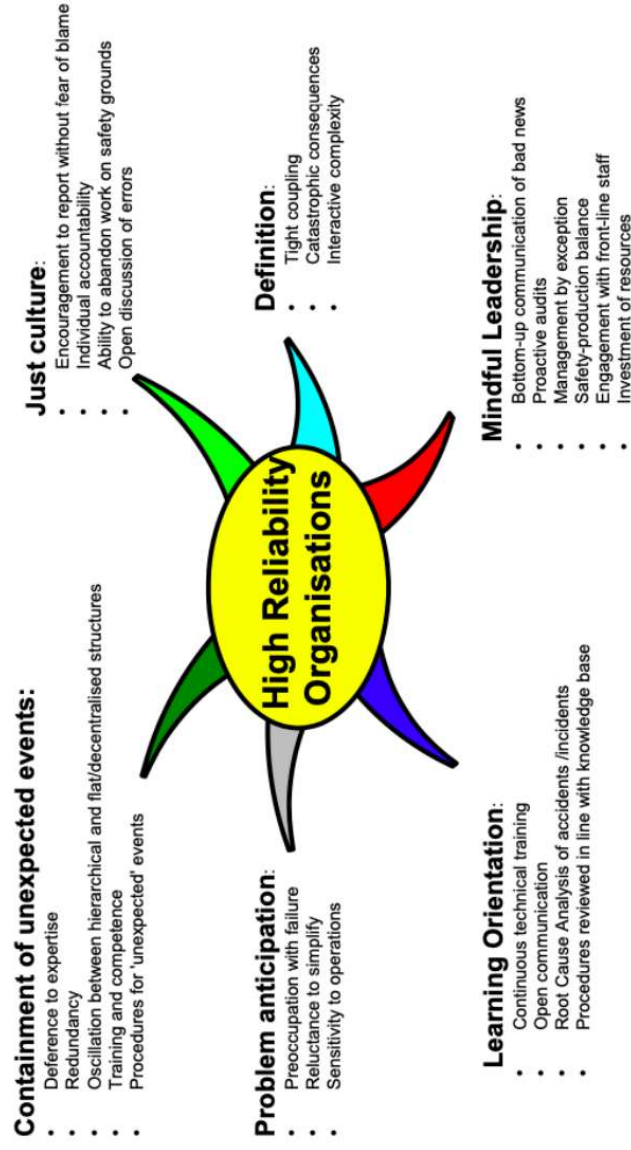
The Problem - Changing Speed of IT Deployment



COMPANY	DEPLOY FREQUENCY	DEPLOY LEAD TIME	RELIABILITY	CUSTOMER RESPONSIVENESS
Amazon	23,000/day	minutes	high	high
Google	5,500/day	minutes	high	high
Netflix	500/day	minutes	high	high
Facebook	1/day	minutes	high	high
Twitter	3/week	minutes	high	high
Typical enterprise	once every 9 months	months or quarters	low/medium	low/medium

	High Performers	Medium Performers	Low Performers
Deployment frequency	1,460 deploys yearly	32 deploys yearly	7 deploys yearly
Change failure rate	75%	38%	23.5%
Mean time to recover	1 hour	24 hours	24 hours*
Outage cost	\$500,000/hr	\$500,000/hr	\$500,000/hr
Total cost of downtime per year	\$54.75M	\$145.92M	\$19.74M
Downtime cost per deployment	\$37.5K	\$4.56M	\$2.82M

System-Of-Systems



Role of SRE in Driving Organizational Success

Improving System Reliability:

- Implementing SLIs and SLOs to measure and maintain reliability targets.
- Using error budgets to prioritize reliability improvements versus feature development.

Scalability Through Automation:

- Infrastructure as Code (IaC) to manage resources at scale (Terraform, Ansible).
- Self-healing systems using automation for fault recovery.

Operational Efficiency:

- Reducing toil through automation (e.g., automated deployment pipelines).
- Proactive incident management using predictive analytics.

Benefits of SRE Practices

High Uptime:

- Building and monitoring robust SLIs for uptime metrics.
- Real-time observability tools (Prometheus, Grafana) for proactive monitoring.

Optimized Costs:

- Reducing downtime costs by improving MTTR and MTBF.
- Using capacity planning to prevent over-provisioning of resources.

Increased Revenue:

- Faster page load times leading to better customer retention.
- Scaling infrastructure for peak traffic scenarios (e.g., Black Friday).

Achieving Operational Excellence and Innovation

Balancing Operational Excellence and Innovation:

- Error budgets as a framework for managing risk and innovation.
- Continuous feedback loops to improve system performance and development practices.

Long-Term Reliability:

- Establishing SRE maturity models for progressive improvements.
- Emphasizing post-mortems for iterative learning and preventing incident recurrence.

Summary

Role of SRE: Enables system reliability, scalability, and efficiency throughout the service lifecycle.

Key Practices:

- Defined SLIs and SLOs to measure and maintain reliability targets.
- Error budgets to balance operational excellence and innovation.
- Automation to reduce toil and improve scalability.

Benefits:

- High uptime for better user experience.
- Optimized costs through reduced downtime and efficient resource management.
- Increased revenue by enhancing system performance and reliability.

Long-Term Success: Achieving a seamless balance between innovation and reliability by adopting SRE maturity models and continuous learning from post-mortems.

“SRE is not just about reliability—it’s about driving business success through innovation and operational excellence.”

Aha slides

- What are key SLIs and SLOs you would define for a food delivery app?
- What words come to mind when you think of an error budget and its importance?
- What tasks would you automate to ensure system reliability?
- What are the key metrics to monitor for ensuring a website's availability?
- What steps would you take to investigate a sudden spike in latency?
- What key elements should be included in a blameless post-mortem?
- What are potential failure scenarios you would simulate in a test environment?
- What observability tools do you use or know, and how do they help?
- What terms define the importance of SLAs for customer-facing applications?
- What are some ways to improve the reliability of a new service?

Thank You!
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